Open-Source Health Information Technology: A Case Study of Electronic Medical Records

Abstract

Objectives: Open-source software (OSS) has achieved widespread adoption in many domains such as server operating systems, enterprise tools, and databases. We review the status of open-source in healthcare and discuss the potential of open-source to resolve some of the challenges surrounding the wide adoption of Health IT in North America. Specifically, we address aspects that are unique to using open-source in ambulatory care. We present the case of a Canadian open-source electronic medical record system (EMR) named OSCAR that is widely adopted by primary care units and family physicians in Canada.

Target audience: health practitioners, health technology managers and policy makers.

Methods: A case study based on semi-structured interviews with various stakeholders of the EMR including users (medical and staff), developers, and service providers. In addition, we assess arguments presented in the literature for and against open-source software in healthcare in light of the OSCAR EMR.

Results and conclusions: Open-source development provides a unique platform that enlists contributions from various stakeholders toward creating a common good. As evident from the OSCAR case, the dynamics of OSS development are potential solutions for the low adoption of technology in healthcare. In particular the low cost of acquisition and maintenance, the high degree of customizability, and the community of users are important advantages of OSS EMR. On the other hand, the lack of understanding of OSS model and the unavailability of trusty support providers are unique challenges facing OSS EMR.

Keywords

Open-source software, open innovation, electronic medical record, health records, health IT

Introduction

Open-source software (OSS) development has gained momentum and emerged as a serious competitor to traditional proprietary software. Adoption of OSS has gained a strong foothold in public sector and academia (1), however in healthcare, open-source has yet to gain wide acceptance. Commercial adoption of OSS is still not fully understood because of the complex and novel issues surrounding open-source (2). For example, several open-source healthcare IT (HIT) systems developed in North America are still not widely adopted (3, 4). There is an increasing interest in OSS in healthcare. The unique characteristics of OSS represent potential solutions for hurdles associated with adoption of EMR systems (5). Such characteristics include: low cost of acquisition and maintenance, a greater possibility of customization, and a lower exposure to vendor failure and product termination. At the same time, there are distinct challenges that face open-source health IT. They include the lack of familiarity with the open-source development model, a poor integration with existing vendor-based health systems, a fragmented development effort, a limited number of support providers, an absence of a clear software development roadmap, and the lobbying and marketing power of commercial vendors of HIT (6).

This paper examines a specific case of open-source HIT in ambulatory care. OSCAR, a Canadian open-source Electronic Medical Record (EMR), has been in existence for a decade and is currently used by over 1,500 Canadian doctors to follow over a million patients. The OSCAR EMR is freely available open-source software and is gaining market share against commercial products typically costing thousands of dollars per year. Furthermore, contrary to other open-source projects, OSCAR is embedded in a community of doctors rather than a community of programmers. The community is active both face to face via user meetings and online via discussion forums. Given the complexity of such software, the mission-critical nature of patient records, and the fact that the vast majority of medical users are not computer savvy, this success is unusual and significant. Following this introduction, we outline the obstacles impeding the adoption of HIT and the arguments for and against open-source in healthcare. We then present the case study of OSCAR and reflect on how OSCAR may realize the benefits and respond to the challenges of open-source HIT.

Challenges of Wide Adoption of Health IT

The use of technology in healthcare is associated with many advantages such as cutting down healthcare costs and improving the quality of care. It is estimated that, by improving health care efficiency and safety, the widespread adoption of electronic medical record (EMR) in the United States can save more than $77-$81 billion annually. In ambulatory care, EMR systems are found to reduce the number of unnecessary or marginally productive visits by 9% (7). In hospitals, the implementation of integrated physician order entry (CPOE) and ICU Physician Staffing (IPS) is associated with improving patient safety and reducing cost. Such systems could save over 65,000 lives, prevent over 907,000 serious medication errors, and save approximately $9.7 billion annually (8, 9).

In spite their advantages, HIT are not widely adopted in North America compared to other organizational information systems such as enterprise resource planning (ERP) systems. For example, in a 2008 national survey, only 4% of physicians in the United States reported having a fully functional EMR system and 13% reported a basic system (10). Only 1.5% of US hospitals reported having a comprehensive EMR system and 7.6% had a basic system. Computerized physician order entry (CPOE) systems were implemented in 17% of US hospitals (11). During the last five years, the adoption of health records grew rapidly in the US and Canada, in part because of government push and support (12). As of 2013, only 44% of US hospitals have a rudimentary EMR and 5% have a full-fledge one (13).

There are many challenges that face the widespread adoption of IT in healthcare (10, 11). The decision to adopt HIT depends on the characteristics of the innovation, the characteristics of the organization, and the characteristics of the external environment (14). Those characteristics along with interconnections with economics and politics, make introducing change difficult in healthcare (15). Barriers against wide adoption of HIT exist from both providers and consumers' perspectives. Providers of healthcare include both large institutions such as hospitals and small providers such as independent physicians and small clinics. Several issues must be addressed to promote successful adoption of electronic medical records (EMR) in clinics: financing, interpretability, standardization, and connectivity of clinical information systems, training and technical support (16). Those issues are similar for hospitals but on a more complex and larger scale where various stakeholders are involved in the process (17).

Cost

Cost is the primary barrier to adopting HIT systems (12). A recent estimate of the cost of purchasing an EMR in the US is $15,000 to $50,000 per physician (18). Operating costs may reach $20,000 per year (19). Costs soar when the system is implemented in a larger organization. For example, a computerized physician order entry (CPOE) project took three years to complete and cost $18 million dollars in 1995 money (20). Nationwide implementations require a huge investment to be realized. For example of, the Quebec Health Record had an initial budget of $562 million (21, 22).

Complexity

HIT systems should be user-friendly and intuitive in order to reach wide acceptance and save on learning time, an important factor for adoption. As an example, the transition to an EMR system in one clinic slowed the work of physicians and resulted in a 10% to 20% reduction in productivity for a period of months or more (23). Other research reports major slowdowns at the initial stages of adoption (16). Customizability is also required because of the richness and diversity of medical practices. HIT systems should disguise their complexity to their users. This can be achieved via careful design of the user interface and allow for a high degree of customization in order to match the needs of different groups of users (24).

Interoperability

Because patients need to interact with multiple health workers and insurers in multiple health units, patients' data should be portable and compatible with the different systems used in different sites. Governments' involvement in promoting open standards for healthcare is essential for the success of building interoperable systems. Some countries such as the UK and Taiwan chose to implement an EMR system to be used nationwide (25, 26). In North America, however, the trend is toward allowing different competing systems while insuring interoperability of the data. One solution is to store some data centrally and allow access to it by different systems. For example, the Quebec Health Record stores records of patients in Quebec centrally (21). Along with storing some data in shared repositories, different systems by different vendors should use a common protocol for communication such as HL7 (27).

Providers' acceptance

The challenges of health IT go beyond the characteristics of the technology artifact and how best to design an IS that replaces the existing system (28). Adopting information systems in healthcare is not only a question of cost-benefit analysis but also a complicated decision that depends on the unique social factors that are embedded in healthcare (29). Careful attention should be paid to social and cultural norms in the medical institution for a successful implementation (30, 31). In addition, implementing IT in the hospital or the clinic brings change into a long established profession in time-honored institutions. Time required to learn something novel, fear of potential lawsuits, risk of data openings, fear of automation and de-skilling, and poor track record of health IT are among many obstacles that may arise when computerization of patients' records is introduced (32). In particular, implementing EMR in clinics should not to impose any additional requirement on the physicians and minimize impact on physician-patient interaction (16).

Consumers' acceptance

Most HIT issues are provider-side issues because health data including patients' records is traditionally stored at the provider-side and accessed and processed solely by providers. Computerization of health data removes this restriction as data can be stored in different locations and shared among providers, insurance companies, government agencies, and patients. One big issue is privacy as patients may have concerns when their health data is collected and processed by HIT. Concerns of privacy may impede the diffusion of an EMR system that has been demonstrated to reduce medical errors (33). Other issues include data ownership, possession, and disclosure of health data. New legislations and legal structures are required to deal with those issues (34).

Those recurring challenges provide an opportunity for new models of technology development to prove their viability (35, 36). Alternative models of innovation and technology production such as open innovation and open-source development can solve problems that traditional models could not (37, 38). Whether the success of OSS can translate to healthcare is still an open question. The paper contributes to an answer by presenting theoretical arguments for and against OSS in healthcare and by then examining the case of OSCAR EMR in ambulatory care.

Advantages and Disadvantages of Open-Source in Healthcare

OSS combines aspects of private investment and common action production models. This new combination requires fundamental revisions to models of innovation (39). In healthcare, OSS has much strength and promises multiple opportunities. At the same time, OSS suffers from several weaknesses and threats (40). Strength and opportunities include: low acquisition, installation and maintenance costs, community support, shared development costs, less vulnerability to vendor vulnerability or product termination, and a greater opportunity for customization. Weaknesses and threats include: increased costs for early adopters, functional limitations, lack of standards and interoperability, limited technical support and lack of public trust concerning security and privacy (40).

The main benefit of open-source HIT is its lower acquisition and maintenance cost compared to proprietary systems. As a common good, the cost of developing and supporting OSS is shared by different stakeholders. In addition, OSS subsidizes the initial big acquisition cost over yearly maintenance and support expenses. Moreover, the client is free in choosing support providers. As a result, support cost for OSS is significantly lower than that of proprietary software (41). The bulk of the initial costs come from hardware, installation, training and support. Local IT expertise could realize supporting tasks without being tied to a supporting company. On the other hand, it is suggested that the total cost of ownership (TCO) is higher in OSS HIT than in proprietary software (6). However, TCO calculation is a complex computation that depends highly on many contextual factors (42, 43). Furthermore, the quality of support is as important as its cost. Currently, there is a lack of big companies that back open-source HIT. In contrast to OSS HIT, OSS Enterprise systems have well-known companies supporting them (e.g. RedHat supporting Linux). The reliance on small companies to support open-source HIT is an issue for health institutions that are looking for a durable and trusty source of support. In most OSS projects, the community of users also plays a vital role in providing help and support. In contrast to proprietary systems where the vendor is in charge of providing support, users of open-source HIT rely on each other to acquire and share knowledge about the system. Mature open-source communities can devote more time and resources to solve arising issues and problems than do traditional organizations (44).

OSS promises to eliminate the risk of vendor locks-in and results in a stronger position for the purchaser and a lower cost of service (5). OSS development relies on pooling and organizing distributed talents and skills. The free distribution of source code, in addition to the availability of distributed expertise eliminates the lock-in situation that happens with proprietary systems. Moreover, the decentralized development and support model also decreases the risk of vendor failure or abrupt product termination. However, it is worth mentioning that many OSS HIT projects do not get enough traction and become abandoned overtime (4). Successful OSS HIT projects seem to have commercial and government sponsors (45). In addition, most health institutions have already adopted HIT. Commercial vendor lock already limits the possibility of future OSS adoption in many health institutions. In a survey to public and non-for-profit US hospitals, all instances of OSS HIT adoption were not switchers from a propriety system to OSS but rather from paper to OSS HIT (46). Moreover, aversion to OSS in hospitals does not extend to non-health OSS such as operating systems, databases, web browsers and application servers (47).

Interoperability is highly desired in a highly regulated sector such as healthcare. OSS favors open standards and protocols to insure that efforts of multiple parties can integrate. This openness in term of exposing the code has internal and external advantages. Internally, the free and unrestricted access to the code allows for more flexibility in changing the system and customizing it. Externally, open-source systems tend to comply with standard programming interfaces and open architectures such as the Health Level 7 (HL7) protocol in the medical context. OSS HIT should be designed with open standards and specifications in mind (48). This insures higher interoperability and easier connection to other health systems. However, network effects may favor existing proprietary HIT. A particular proprietary HIT may appear to offer a de facto standard for interoperability although other OSS may be as interoperable as their proprietary counterpart (49). In addition, open-source HIT systems may not integrate well with health systems deployed at medical institutions such as hospitals and pharmacies. Large-scale systems are typically developed by big vendors and are designed to integrate well with complementary products from the same vendor. Finally, some argue that by being open, OSS are less secure because an attacker can access the source code and learn about the inner working of the system in order to exploit vulnerabilities. However, this argument is controversial because vulnerabilities can be quickly fixed since the source code is accessible by many people. Finally, open-source HIT may lack the availability of coded data that

may be proprietary or strictly licensed including databases of diseases, drugs, procedures, and billing codes. Such coded data is necessary to comply with standards and to limit the amount and variation of free text entry in the system. While some of these databases are standardized and available (e.g. ICD-9 disease database), some of them require extensive effort to compile and maintain (e.g. drug databases) and some depend on the local context (e.g. billing and procedure codes). Fortunately, there is a recent movement towards open access platforms and databases (50, 51).

Environmental obstacles may impede the adoption of open-source. Interviews with government officials and health managers indicate that barriers to adopting OSS HIT are more organizational than technical (46, 52). The open-source development model is not properly understood by many stakeholders including medical practitioners. In particular, the lack of internal knowledge about OSS as well as the conservative nature of healthcare and political pressures on policy makers limit the potential of adopting OSS in health institutions (52). Also, there remains a widespread misconception that because open-source HIT is cheaper than commercial software, it must be of lower quality. This lack of understanding, along with the significant power of lobby groups that represent commercial vendors, contributes to weak support for open-source health IT initiatives (6). Awareness of other health organizations using OSS and collaboration with open-source communities are major facilitators in adopting OSS HIT in hospitals (49).

To summarize, arguments for open-source in healthcare include: economic factors where the economy of scale works in favor of large open-source projects with large communities (6), personal factors because members of open-source communities have personal motivation such as peer recognition to be involved in the community (53, 54), and social factors because open-source communities tend to be fluid favoring creativity and innovation (55). Notwithstanding these intrinsic factors, the widespread adoption of infrastructure OSS such as Linux, MySQL and TomCat stems from the response to consumer demand, cost, and competition among vendors (Oracle & IBM) to dominate the OSS market (56). These factors require a critical mass and the support and sponsorship of various stakeholders. Indeed, many successful OSS HIT projects have commercial sponsors (45). As major payers of health care bills in many countries, governments should take the lead in sponsoring OSS in healthcare (45). Such support is important for the success of OSS HIT (57).

Methods

We examine OSCAR, a Canadian open-source EMR. In order to obtain a comprehensive understanding of the different aspects of OSCAR and the community in charge of developing it, we have conducted semi-structured interviews from multiple locations: a clinic that adopted OSCAR to replace its paper records, the main development team, several software companies providing support to OSCAR, and the user community. Table 1 summarizes the data collection process.

1. A clinic adopting OSCAR: we interviewed doctors, nurses, managers and staff at a medium-size clinic that switched from paper charts to OSCAR in 2008. The first author followed the switch since 2009 by attending weekly meetings in the clinic and observing discussions around the EMR.
2. OSCAR development team: we interviewed programmers and developers working in the main development team at McMaster University. We talked to four lead developers and other interns in charge of main development and coding of OSCAR.
3. OSCAR service providers (OSPs): OSPs are commercial companies that provide OSCAR installation, support and training services to their clients. We had the chance to interview four OSPs. In addition, we have a good relationship with an OSP based in Montreal that provided us with information about OSCAR growth in the city and the various clients' cases.
4. OSCAR user community: we attended six OSCAR user community meetings in Montreal, Toronto, Hamilton, and Vancouver in 2011 and 2012. We documented the discussed topics and followed the interests of different stakeholders. We conducted nine interviews with members of the community.

The interviews were recorded, transcribed verbatim, and analyzed using a qualitative data analysis program. The goal of the analysis is to cover perspectives of various stakeholders about OSCAR. For example, interviews with medical practitioners and community members focused on issues surrounding the adoption of the EMR and the difference open-source makes for health professionals. On the other hand, interviews with developers and service providers focused on the development process and the business model associated with OSCAR. Table 2 provides the questions used in the interviews.

<Table 1 approximately here> <Table 2 approximately here>

The OSCAR Project

OSCAR (Open Source Clinical Application Resource) is an Electronic Medical Record (EMR) system developed at the Department of Family Medicine at McMaster University and deployed in multiple clinics in Canada and worldwide (http://www.oscarcanada.org/). OSCAR EMR features a web-based interface that allows multiple users to access the system simultaneously through different browsers and platforms. The web interface also allows access to the system over the Internet through a secure connection. The number of sites adopting OSCAR for their practices has steadily increased since its release in 2002. Today OSCAR is used by more than 1,500 sites that serve more than one million patient records. OSCAR gained substantial market share in major Canadian provinces reaching third place in Ontario in 2012. In addition to independent clinics, OSCAR is the EMR of choice of many teaching faculties in Canada.

Hardware and Software Architecture

As open-source software, OSCAR is capable to run on different hardware platforms and configurations. The typical hardware configuration consists of a server with data duplication mechanisms such as RAID-1 or a secondary server that is synced with the main one. Clients may be at any number of terminals such as laptops and desktops that are connected to the server via a local network or the Internet if the server has an internet connection (Figure 1). Many other configurations are possible as well. The choice of a platform and a configuration depends on the requirements of the site.

OSCAR is built around a client-server model where different clients access a central server over a local network or the Internet. OSCAR is written in server-side Java. It features a web interface written in HTML, JavaScript, Java Servlets and Java Server Pages Scripts. The default backend database is MySQL. The default installation of OSCAR runs on the top of the Apache Web server with Java Runtime Environment enabled via the Tomcat application server. Ubuntu Linux is the preferred operating system although OSCAR runs on other distributions of Linux, Windows and MacOS. OSCAR is modular software composed of various modules that implement different functions.

<Figure 1 approximately here>

Functionality

OSCAR provides a wide spectrum of clinical, administrative and support functions that are comparable to offerings of other EMRs (Figure 2). Clinical functions include electronic charts, prescriptions, labs, consultations, disease registry, measurements, preventions, allergies, and clinical forms. Administrative functions consist of billing, patient management (patient registration and demographics), appointment and schedule management, and an electronic document repository (Figure 3). Support functions include system functions such as user management and administrative settings. In addition, they include OSCAR report, a reporting tool that generates various types of reports. OSCAR messenger is an email-like communication module that allows users to send and receive messages inside OSCAR, and OSCAR tickler is a patient-centric reminder and alarm tool. Finally, OSCAR allows the rapid development of electronic medical forms to extend the built-in functions and forms. An e-form is simply an HTML form with additional tags that allow it to communicate with the database pulling and posting information there. Once an e-form is uploaded, the e-form module parses it and creates a table in the DB with corresponding fields. This translation is seamless, and the only requirement to write e-form is basic knowledge of HTML in addition to learning the specific OSCAR tags.

<Figure 2 approximately here>

Development ecosystem

OSCAR is based on a community driven development model where multiple individuals and institutions contribute to project development. The contributors include the main development team at McMaster University, large institutions such as McGill University, independent clinics and doctors, and independent service providers interested in the business opportunity. The core development team, located at McMaster University, is responsible for maintaining the code repository and overseeing other members' contributions (58).

Funding

The OSCAR project receives funding from multiple sources. Government is the main grantor. In 2012 OSCAR received a grant of $5.8M from the Federal Economic Development grant for their Connected Health and Wellness Project (CHWP) initiative (59). The goal of CHWP is to develop a new people-centred and technology-enabled system that makes it possible for patients to keep health information from different hospitals or clinics in one place, and lets them decide who they trust to see that information. OSCAR also receives funding from provincial agencies such as Ontario Medical Association's Ontario Primary Care Reform EMR incentive funding. At the local level, OSCAR benefits from McMaster University's commitment to the project and from various resources afforded by the university and affiliated medical institution. Finally, because OSCAR is certified in Ontario, OSCAR's users are incentivised to adopt OSCAR by OntarioMD funding for eligible EMRs (60).

In addition to grants, OSCAR benefits from private providers building their business around installing, supporting and maintaining the EMR (61). OSCAR McMaster certifies third party providers, also called OSCAR Service Providers (OSP), for a yearly fee. In return OSPs receive training and certification. Finally, OSCAR receives funds from its users and developers. In addition to dedicating time and effort to the project, OSCAR users can donate money to the OSCAR project. In 2013, 50% of funding came from McMaster University, 45% from other grants and 5% from the certification program. OSCAR's goal for the coming years is to increase the received funds from the certification program and reduce its dependency on grants.

Community involvement

OSCAR has a vibrant community of medical users and professional developers. Most users are members of OCUS (OSCAR Canada Users Society) a non-for-profit organization that represents the interests of the user community in the OSCAR project. The community maintains an active mailing list with an average of seven messages per day. The community holds regular national and regional meetings where members can meet face to face and exchange their ideas and expertise. The community also contributes to the OSCAR project. These contributions range from simple additions such as templates and medical forms to sophisticated enhancements and plug-ins.

OSCAR Service Providers (OSPs)

OSCAR Service Providers (OSP) are independent computer and software companies who provide installation, training and support services to OSCAR users. OSPs are also encouraged to contribute fixes and improvements to the OSCAR project. Because OSCAR is open-source, OSPs do not have the right to sell licenses to use the software. However, they can sell related services such as installation, training and support. They can also sell software extensions built around OSCAR. Currently there are over fifteen OSPs in Canada and their number is growing rapidly (58).

Does Oscar Realize the Benefits and Respond to the Challenges of Open-Source?

Cost of OSCAR

The high cost of EMRs is perhaps the biggest challenge to their wide adoption in small and medium size clinics. Based on the values of academic research and knowledge sharing, OSCAR is not designed to profit the developers. However, value generated processes are required to enlist interested third party developers and service providers. The availability of service providers is a must to gain a critical mass of healthcare users to adopt and continually use OSCAR (62). Although the OSCAR itself comes at no cost, there is a cost associated with getting support as well as acquiring the hardware infrastructure and maintaining it. Therefore, the cost depends on the type of installation and the size of the site. Typical costs for a small clinic as provided by Indivica, one OSCAR support provider based in Ontario, are provided in Table 3 (63).

<Table 3 approximately here>

We could not obtain figures from competing commercial vendors who do not publish cost of their systems and also declined several requests to disclose the cost of purchasing and supporting their systems. However, talking to two clinics that have a proprietary EMR we were able to obtain the following approximate numbers: $7,000 yearly license fee per doctor, $1,500 yearly support fee per doctor, and $140/hour for training and extra tasks.

Support and Maintenance

There are many IT consultants and several companies (OSPs) that specialize in providing OSCAR installation, maintenance and support in Canada. In addition, the user community online list is an excellent resource for support and help over a wide range of requests and questions. OSCAR's active user community, the availability of independent companies supporting it, and its affiliation with large teaching institutions such as McMaster University and McGill University are all factors that contribute to the increasing visibility, recognition and adoption of OSCAR in Canada.

Community Contribution

The user community contributes actively to OSCAR. In addition to collaborating and communicating with the core development team, the user community contributes a range of additions and improvements to OSCAR. We outline here two examples of community contributions:

1. The e-form generator is a module that allows OSCAR users to quickly build new medical e-forms by scanning paper forms (e.g., x-ray requisition) and overlaying data placeholders such as textboxes and checkboxes over the scanned form (Figure 3). The generator then generates HTML code that can be added to OSCAR as an e-form. The generator cuts down development time and reduces the difficulty of writing new forms considerably. Furthermore, many users share the e-forms they develop (http://oscarcanada.org/oscar-users/emr-resource/eform/eform-examples). As a result, a wide variety of medical forms is freely available to OSCAR users. This is a killer feature for many OSCAR users especially when compared to commercial EMRs that charge an hourly rate for developing e-forms.
2. The OSCAR manual (http://oscarmanual.org/) utilizes wiki technology that allows users to contribute by creating and modifying the content of the manual. The result is a collaborative product of both users and developers available for the public under Creative Commons License. The spirit of collaboration and sharing in the OSCAR project therefore extends beyond source code to include the documentation of the EMR.

Open Standards & Ports

OSCAR complies with different standards and certifications. OSCAR meets the requirements of open source as defined by the Open Source Initiative, and the process of development complies with ISO 9000 standards of software development. Furthermore, OSCAR has passed provincial EMR certification in Ontario and Quebec. OSCAR has passed all rounds of Conformance testing in Ontario, including the most recent and rigorous CMS Specification v3.0 from OntarioMD. OSCAR can import and export clinical data and labs via HL7. OSCAR comes with English and French language support out of the box.

Availability of Coded Data

OSCAR includes a variety of coded data databases including diseases, drugs, and billing. OSCAR supports the International Statistical Classification of Diseases and Related Health Problems (ICD) with versions nine and ten. OSCAR also supports adding new diseases to the disease registry. OSCAR includes two billing databases and modules for Ontario and British Columbia. The administrative settings allow uploading billing codes and modifying the existing ones. Finally, OSCAR integrates with drugref.org a drug database containing the Health Canada drug database and a drug-drug interaction reference based on a collaborative open content database. The database contains around 25,000 products which companies have notified Health Canada as being marketed (64).

Security and Privacy

OSCAR takes security and privacy of medical information very seriously. There are three layers of security in order to insure that patients' confidential medical data is safe. The first layer is hardware security that makes sure that even if the server is accessed by intruders the data stored on it cannot be read. Hard disk and database encryption are employed. In addition, all automatic backups that are done by OSCAR are encrypted so that the data will be useless if accessed accidently by intruders. The second layer of security is connection security that takes care of securing communication between the server and the clients accessing the server over local network or via the Internet. The third layer of security is application-level security. OSCAR access is secured via a two-password login process. One password is set by the user and the second is set by the system administrator.

Results and Discussion

We summarize here the results of our interviews with clinicians in the studied clinic, OSCAR developers, service providers and community members. Selected quotes pertinent to various issues and different groups are listed in Table 4.

<Table 4 approximately here>

A Clinic adopting OSCAR

We interviewed medical practitioners in a medium-size walk-in clinic in Montreal. The clinic is affiliated with the medical center of a large university and serves approximately 30,000 patients per year. Its staff complement includes 38 physicians, 15 nurses, and 10 clerks. Embedded in the health network of the university, the clinic receives and refers patients to other health centers affiliated with the university health network.

In 2008 the clinic decided to switch to EMR from paper charts as the paper charts were becoming harder to maintain. Paper charts occupied valuable office space. In addition, exchanging paper charts with the hospital via the courier service has become inefficient with increasing number of patients visiting the clinic. In the beginning, there were concerns about OSCAR, in particular its maturity and security. Privacy of medical data is a big concern. The fact that OSCAR passed certification requirements of several provinces and that it has been adopted by many other clinics eased those concerns. Shortly after installing the EMR, the clinic management invited OSCAR developers from McMaster to train the staff at the clinic. Those training sessions were short and served just as an introduction to the system. Afterwards, medical practitioners had to use the system, learn its various features and solve arising problems. Initial experience was not great. Switching from a paper system to OSCAR was a big change. In addition, OSCAR's interface was not intuitive for most clinicians and had a learning curve. The fact that clinicians did not have to contribute financially for OSCAR nor switch immediately helped in reducing the negative consequences of the initial clumsy experience.

In order to solve the issues and problems arising from using OSCAR, meeting sessions to discuss OSCAR were scheduled weekly. The weekly meetings turned to be popular among staff and clinicians.

Attendants were mostly physicians in addition to nurses, technicians, and other staff members. During those meetings attendants exchanged their experience and taught each other. Meetings were opportunities for learning and sharing. A great deal of time during the meetings was dedicated to problem solving. In the absence of a traditional source for technical support such as a technical support provider or a technician on site, users had to find their own solutions. Some problems could not be solved in-house and help was often sought from users in other sites by sending messages to the community mail list. At many other times creative solutions emerged via exchanging ideas, trial an error and collaboratively working out a solution. Some physicians got excited and became champions of OSCAR. They attended most meetings and participated actively. They also supported their colleagues and promoted OSCAR among them. Champions also interacted with other users in the online mail list conveying other colleagues' questions to the community and bringing back answers and suggestions.

Overtime many required changes and customizations emerged for several reasons including institutional, clinical and administrative needs. For example, OSCAR had to comply with provincial legislation and had to be available in both French and English languages. Other customizations included adding clinical forms and templates, setting up user accounts and rules in addition to many others. Faced with the need to customize the EMR, the clinic continued to seek help from the community and other available sources. They had also to improvise their own solutions in many other cases. OSCAR user community provided extensive help and support via the online mail list. Much of the time during the weekly meetings was dedicated to solving problems.

OSCAR developers and service providers

The OSCAR development team at McMaster University is in charge of developing and maintaining the core functions of OSCAR. In addition, the team oversees the new feature and change requests by OSCAR users and reviews the code contributed to the project by other developers and service providers. In addition, the McMaster team ensures that OSCAR development complies with ISO standards and that OSCAR remains certified by OntarioMD. The role of other developers and OSPs is geared toward getting OSCAR to operate in a specific regions, health networks, or clinics in Canada. In particular, OSPs provide quality implementation, training and support services for the OSCAR clients. OSPs also customize

OSCAR to the particularities of their client sites and implement new required features. For example, the provincial requirements and specifications for EMRs in Quebec differ greatly from those in Ontario where OSCAR is developed. One OSP in Quebec took on the duty of certifying OSCAR. The process required several tasks such as integrating OSCAR within the provincial health network for exchanging patients' information and implementing billing functions. The OSP is planning to submit its contribution to OSCAR McMaster team that will evaluate the changes and integrate them into the mainstream OSCAR distribution. The fact that the McMaster development team holds a tight grip on OSCAR distribution causes some tension with OSPs who want more autonomy and control in pursuing their agendas. Finally, the ability of doctors to support themselves and bypass service providers is also a thorny issue for OSPs.

OSCAR user community

The OSCAR community is perhaps the most important feature of OSCAR after its zero price. The community plays an important rule to attract new users and also support them. Community members emphasized the role and added value of the community to the OSCAR project. The value of the user community stems from various activities besides online interaction. Those activities include training and demonstrating OSCAR, organizing meetings, and supporting other users. In addition, some advanced users in the community contribute code while less tech savvy users contribute by reporting bugs, testing new features and writing medical forms. Finally, the community adds a personal dimension of being an OSCAR user. Community members have a sense of ownership and control of OSCAR and form a bond that unites them in helping themselves and advancing the project.

By affording rapid development of plug-in medical templates and forms OSCAR helps its users customize the system without much technical knowledge. E-forms became the place where OSCAR gets customized by its users. The beauty of e-forms is easy development, unneeded special privileges to add or modify e-forms, and the ability to share e-forms with other users. Doctors managed to implement very advanced e-forms including e-forms for end of life care plan, therapeutic drug monitoring, personal health risk assessment, mental health tool kit, Chronic Kidney Disease (CKD) advisor, rich text editing and other e-forms that implement decision making functionalities. Furthermore, these e-forms are shared on the community website for other users to download, modify and use. In addition to developing e-forms by

writing HTML code, the e-form generator, a community developed module, provides an easy WYSISYG tool to develop e-forms based on scanned medical forms. The ease of developing e-forms in OSCAR empowered doctors and other users who do not have programming skills to take on themselves to customize OSCAR in their clinics without relying on OSPs. One doctor noted “who needs a service provider anymore?”

< Figure 3 approximately here>

Conclusions

The use of technology in healthcare is associated with many advantages such as cutting down healthcare costs and improving the quality of care (65). Despite such advantages, the rate of adoption of Health IT in North America is much lower compared to other developed countries (10). Alternative models of technology innovation, software development and service provision are needed to address the issues challenging the wide adoption of health IT. Open-source has been identified as an alternative production model for creating goods and services that combine aspects of private investment and common action production models (66).

Open-source software and its associated development model have been advocated as potential solutions for the low adoption of health IT (6, 67). The unique features of open-source software may alleviate some of the obstacles associated with proprietary systems. Such features include: low cost of acquisition and maintenance, a greater possibility of customization, and lower exposure to vendor failure or product termination. At the same time, there are exclusive challenges that face open-source health IT. They include the lack of familiarity with the open-source development model, the lack of integration with existing vendor-based hospital systems, a fragmented development effort, a limited number of firms that support installation and training, the absence of clear software development roadmap, and the lobbying and marketing power of commercial vendors of healthcare software (6).

OSCAR EMR demonstrates the promise of OSS in primary care. The main advantage of OSCAR over proprietary EMRs is the reduced cost of acquisition and maintenance. In terms of complexity,

OSCAR has a considerable learning curve. However, this training stage is facilitated by the many resources available online and also by the existence of a community of enthusiastic users. Moreover, a huge collection of downloadable e-forms and templates is available online making customization a very easy process. Finally, OSCAR favors open standards and interfaces including support for HL7, ICD-9 & 10, Canadian drug database, and billing codes. More effort is required to integrate OSCAR with legacy HIT systems of hospitals and pharmacies.

As open-source software, OSCAR offers an accessible and open platform for various stakeholders. While primarily developed and maintained by a team of developers at McMaster University, OSCAR enlists contributions from its users' community. The community represents the interests of users in the project. Communication between users and developers online via the mailing list and during user group meetings ensures that OSCAR stays aligned to medical users' needs. In addition, the user community allows users to share their knowledge and expertise creating a space for obtaining support. Moreover, the user community contributes non-trivial extensions to OSCAR such as e-forms, templates, the e-form generator and the OSCAR manual. In the studied clinic, customizability of OSCAR was essential for its acceptance. Customized forms matched clinic paper forms and customized templates helped to match work practices. Customized templates for consultations and referrals improved the external communication with other clinics and specialists leading to better consultation and referral practices. Finally, allowing qualified commercial businesses to provide support and training services ensures that OSCAR reaches beyond tech savvy users to all medical practitioners accustomed to the traditional model of software acquisition.

References

1. Foster DM. Open Source Adoption in Worldwide Governments 2010. Available from: http://software.intel.com/en-us/blogs/2010/01/19/open-source-adoption-in-worldwide-governments/.
2. Glynn E, Fitzgerald B, Exton C, editors. Commercial adoption of open source software: an empirical study2005: IEEE.
3. Faus SA, Sujansky W. Open-Source EHR Systems for Ambulatory Care: A Market Assessment. California HealthCare Foundation, 2008.
4. Sainz de Abajo B, Ballestero AL. Overview of the Most Important Open Source Software: Analysis of the Benefits of OpenMRS, OpenEMR, and VistA. Telemedicine and E-Health Services, Policies, and Applications: Advancements and Developments: IGI Global; 2012. p. 315-46.
5. Kantor GS, Wilson WD, Midgley A. Open-source software and the primary care EMR. Journal of the American Medical Informatics Association. 2003;10(6):616-.
6. Reynolds CJ, Wyatt JC. Open source, open standards, and health care information systems. Journal of medical Internet research. 2011;13(1):e24.
7. Garrido T, Jamieson L, Zhou Y, Wiesenthal A, Liang L. Effect of electronic health records in ambulatory care: retrospective, serial, cross sectional study. Bmj. 2005;330(7491):581.
8. Cutler DM, Feldman NE, Horwitz JR. US adoption of computerized physician order entry systems. Health Affairs. 2005;24(6):1654-63.
9. Hillman J, Given R. Hospital implementation of computerized provider order entry systems: results from the 2003 leapfrog group quality and safety survey. Journal of healthcare information management: JHIM. 2005;19(4):55.
10. DesRoches CM, Campbell EG, Rao SR, Donelan K, Ferris TG, Jha A, et al. Electronic health records in ambulatory care—a national survey of physicians. New England Journal of Medicine. 2008;359(1):50-60.
11. Jha AK, DesRoches CM, Campbell EG, Donelan K, Rao SR, Ferris TG, et al. Use of electronic health records in US hospitals. New England Journal of Medicine. 2009;360(16):1628-38.
12. Wilson JF. Making electronic health records meaningful. Annals of internal medicine. 2009;151(4):293-6.
13. DesRoches CM, Charles D, Furukawa MF, Joshi MS, Kralovec P, Mostashari F, et al. Adoption of electronic health records grows rapidly, but fewer than half of US hospitals had at least a basic system in 2012. Health Affairs. 2013;32(8):1478-85.
14. Marsan J, Paré G. Antecedents of open source software adoption in health care organizations: A qualitative survey of experts in Canada. International journal of medical informatics. 2013.
15. Romanow D, Sunyoung C, Straub D. Riding the Wave: Past Trends and Future Directions for Health IT Research. MIS quarterly. 2012;36(3):iii-A18.
16. Baron RJ, Fabens EL, Schiffman M, Wolf E. Electronic health records: just around the corner? Or over the cliff? Annals of internal medicine. 2005;143(3):222-6.
17. Mintzberg H. Managing care and cure-up and down, in and out. Health Services Management Research. 2002;15(3):193-206.
18. Blumenthal D, Glaser JP. Information technology comes to medicine. New England Journal of Medicine. 2007;356(24):2527-34.
19. Miller RH, West CE. The value of electronic health records in community health centers: policy implications. Health Affairs. 2007;26(1):206-14.
20. Davidson EJ, Chismar WG. The interaction of institutionally triggered and technology-triggered social structure change: An investigation of computerized physician order entry. MIS quarterly. 2007;31(4):739-58.

21. Castonguay C, Marcotte J, Venne M. Task Force on the Funding of the Health System 2008 [cited 2012 August]. Available from: http://www.groupes.finances.gouv.qc.ca/financementsante/en/groupe/membres.asp.
22. Powers P. Québec's e-Health Journey – Assessing the Path Forward. ElectronicHealthcare. 2012;10(3):e30-e44.
23. Wang SJ, Middleton B, Prosser LA, Bardon CG, Spurr CD, Carchidi PJ, et al. A cost-benefit analysis of electronic medical records in primary care. The American journal of medicine. 2003;114(5):397-403.
24. Tang PC, Lansky D. The missing link: bridging the patient–provider health information gap. Health Affairs. 2005;24(5):1290-5.
25. Robertson A, Cresswell K, Takian A, Petrakaki D, Crowe S, Cornford T, et al. Implementation and adoption of nationwide electronic health records in secondary care in England: qualitative analysis of interim results from a prospective national evaluation. BMJ: British Medical Journal. 2010;341.
26. Liu CT, Yang PT, Yeh YT, Wang BL. The impacts of smart cards on hospital information systems--an investigation of the first phase of the national health insurance smart card project in Taiwan. International journal of medical informatics. 2006;75(2):173-81.
27. Kush RD, Helton E, Rockhold FW, Hardison CD. Electronic health records, medical research, and the Tower of Babel. New England Journal of Medicine. 2008;358(16):1738-40.
28. Sujansky WV. The benefits and challenges of an electronic medical record: much more than a" word-processed" patient chart. Western Journal of Medicine. 1998;169(3):176.
29. Anderson JG. Social, ethical and legal barriers to e-health. International journal of medical informatics. 2007;76(5-6):480-3.
30. Berg M, Bowker G. The multiple bodies of the medical record. The Sociological Quarterly. 1997;38(3):513-37.
31. Rivard S, Lapointe L, Kappos A. An Organizational Culture-Based Theory of Clinical Information Systems Implementation in Hospitals. Journal of the Association for Information Systems. 2011;12(2):123-62.
32. Kaplan B, Harris-Salamone KD. Health IT success and failure: recommendations from literature and an AMIA workshop. Journal of the American Medical Informatics Association. 2009;16(3):291-9.
33. Angst C, Agarwal R. Overcoming personal barriers to adoption when technology enables information to be available to others2006. Available from: http://dx.doi.org/10.2139/ssrn.922075.
34. Hodge Jr JG, Gostin LO, Jacobson PD. Legal issues concerning electronic health information. JAMA: the journal of the American Medical Association. 1999;282(15):1466-71.
35. Tapscott D, Williams AD. Wikinomics: How Mass Collaboration Changes Everything: Portfolio Trade; 2008.
36. Jeppesen LB, Lakhani KR. Marginality and problem-solving effectiveness in broadcast search. Organization science. 2010;21(5):1016-33.
37. Von Hippel E. Democratizing innovation: the MIT Press; 2005.
38. Tushman M, Lakhani KR, Lifshitz-Assaf H. Open Innovation and Organizational Design. Journal of Organization Design. 2012;1(1).
39. Von Hippel E, Von Krogh G. Open source software and the "private-collective" innovation model: Issues for organization science. Organization science. 2003;14(2):209-23.
40. Maglogiannis I. Towards the Adoption of Open Source and Open Access Electronic Health Record Systems. Journal of Healthcare Engineering. 2012;3(1):141-62.
41. Ghosh RA. Licence fees and GDP per capita: The case for open source in developing countries. First Monday. 2003;8(12).
42. Wheeler DA. Why open source software/free software (OSS/FS, FLOSS, or FOSS)? Look at the numbers 2014. Available from: http://www.dwheeler.com/oss_fs_why.html.
43. Fitzgerald B. The transformation of open source software. Mis Quarterly. 2006:587-98.

44. Raymond E. The cathedral and the bazaar. Knowledge, Technology & Policy. 1999;12(3):23-49.
45. Janamanchi B, Katsamakas E, Raghupathi W, Gao W. The State and Profile of Open Source Software Projects in health and medical informatics. International Journal of Medical Informatics. 2009;78(7):457-72.
46. Vest JR, Stephens JH. The use and role of open source software applications in public and not-for-profit hospitals in the United States. Health Care Management Review. 2013;38(4):361-70 10.1097/HMR.0b013e318276f9ed.
47. Munoz-Cornejo G, Seaman CB, Koru AG. An Empirical Investigation into the Adoption of Open Source Software in Hospitals. International Journal of Healthcare Information Systems and Informatics (IJHISI). 2008;3(3):16-37.
48. Cavalini LT, Cook TW. Health Informatics: The Relevance of Open Source and Multilevel Modeling. In: Hissam SA, Russo B, Neto MGD, Kon F, editors. Open Source Systems: Grounding Research. IFIP Advances in Information and Communication Technology. 365. Berlin: Springer-Verlag Berlin; 2011. p. 338-47.
49. Glynn E, Fitzgerald B, Exton C, editors. Commercial adoption of open source software: an empirical study. Empirical Software Engineering, 2005 2005 International Symposium on; 2005 17-18 Nov. 2005.
50. Natter MD, Quan J, Ortiz DM, Bousvaros A, Ilowite NT, Inman CJ, et al. An i2b2-based, generalizable, open source, self-scaling chronic disease registry. Journal of the American Medical Informatics Association. 2012.
51. Kapur T, Pieper S, Whitaker R, Aylward S, Jakab M, Schroeder W, et al. The National Alliance for Medical Image Computing, a roadmap initiative to build a free and open source software infrastructure for translational research in medical image analysis. Journal of the American Medical Informatics Association. 2012;19(2):176-80.
52. Paré G, Wybo MD, Delannoy C. Barriers to open source software adoption in Quebec's health care organizations. Journal of medical systems. 2009;33(1):1-7.
53. Lerner J, Tirole J. Some simple economics of open source. The journal of industrial economics. 2002;50(2):197-234.
54. Wasko MML, Faraj S. Why should I share? Examining social capital and knowledge contribution in electronic networks of practice. MIS quarterly. 2005:35-57.
55. Faraj S, Jarvenpaa SL, Majchrzak A. Knowledge collaboration in online communities. Organization science. 2011;22(5):1224-39.
56. Munoz-Cornejo G. An empirical investigation into the adoption of open source software in hospitals [Ph.D.]. Ann Arbor: University of Maryland, Baltimore County; 2007.
57. Øyri K, Murray PJ. osni.info—Using free/libre/open source software to build a virtual international community for open source nursing informatics. International Journal of Medical Informatics. 2005;74(11–12):937-45.
58. OSCARCanada. OSCAR Support 2012 [cited 2012 September]. Available from: http://www.oscarcanada.org/support.
59. YorkMediaRelations. Connected Health and Wellness Project: A New People-Centred, Technology-Enabled Health & Wellness System 2014 [cited 2014 February]. Available from: http://news.yorku.ca/2012/09/12/connected-health-and-wellness-project-a-new-people-centred-technology-enabled-health-wellness-system/.
60. OntarioMD. EMR Offerings 2014 [cited 2014 February]. Available from: https://www.ontariomd.ca/portal/server.pt/community/emr_offerings/inv0#1.
61. OSCARSupport. OSCAR Support 2014 [cited 2014 February]. Available from: http://oscarcanada.org/support.

62. OSCARManual. About Open Source 2012 [cited 2012 August]. Available from: http://www.oscarmanual.org/introduction/about-open-source.
63. Indivica. What You Need To Consider When You Are Getting Started 2014 [cited 2014 Feb]. Available from: http://indivica.com/?page_id=118.
64. OSCARCanada. Drugref 2012 [cited 2012 September]. Available from: http://www.oscarcanada.org/oscar-users/drugref/index_html.
65. Walker J, Pan E, Johnston D, Adler-Milstein J, Bates DW, Middleton B. The value of health care information exchange and interoperability. Health Affairs. 2005;24:5.
66. Von Hippel E, Von Krogh G. Open source software and the" private-collective" innovation model: Issues for organization science. Organization science. 2003:209-23.
67. Safadi H, Faraj S. The Role of Workarounds during an Opensource Electronic Medical Record System Implementation. 2010.

Table legends

- Table 1: Summary of data collection
- Table 2: Guides for semi-structured interviews
- Table 3: Cost of OSCAR for small clinics
- Table 4: Excerpts from interviews

Figure legends

- Figure 1: Typical OSCAR Installation
- Figure 2: OSCAR electronic patient chart
- Figure 3: Interactive electronic form creation using the e-form generator tool

Funding

This work was supported by The Social Sciences and Humanities Research Council of Canada (SSHRC) grant number 410-2009-1166

Competing interests

Two authors are involved in the OSCAR EMR project. One author initiated the OSCAR project in 2000 and directed the development of the OSCAR system at McMaster University. Another author manages the clinic using OSCAR EMR studied in the paper.

Ethical approval

Not required